Introduction

This project is an analysis on crime in the UK. We have been placed within a real estate company and our task is to know which locations are most/least desirable along with any other useful information that may help our head of sales make informed decisions.
For this task, we have been requested to focus on providing a general idea of crime across regions and a clear link between how the insights generated will assist sales is not included.

Purpose

While crime data across the UK is publicly available information, it is stored in tables and difficult to interpret. As such, the purpose of this task is to perform data preparation and exploratory data analysis with the goal of creating a foundation for which a comprehensive analysis can be done. This task also asks us to select two police forces for further study for the next part of this assignment.

Scope

This project will use crime data from the regions of Thames Valley, Warwickshire, West Midlands, and West Yorkshire.

The two sections of the project have varying scopes. For the data preparation, this section is meant to be more comprehensive as it aims to uncover all foreseeable issues with the data and also acts as a deep dive to consider the possible analyses that are possible with this dataset. Aside from data quality, technical limitations are also considered such as computing power limitations if the dataset is too large.

For the exploratory data analysis, this section is meant to be more broad and is meant to paint a picture of what information the dataset contains and the possible knowledge that could be uncovered through it. This section aims to accomplish this in three ways. The first is to perform multiple types of aggregations such that the different “views” of the data are known. Next is to present distributions of the data through visual mediums such that any areas of interest can be identified and comparisons can be made. Finally, additional dimensions such as time and location data will be used to present time series and map data with the goal of drawing connections beyond aggregations and hinting at more advanced analytics techniques such as predictive analytics

Project Tasks

A simple breakdown of project tasks are as follows:

1. Data Preparation
    a. Data Importing
    b. Data Cleaning
    c. Master Dataset Exporting
2. Exploratory Data Analysis
    a. Data Aggregation
    b. Statistical Analysis
    c. Data Visualisations
        i. Distributions
        ii. Time Series Data
        iii. Map data

Milestones and Deliverables

The following deliverables will be completed by the assigned deadline of Monday 16th of March 23:59:

1. README.md file - Contains key information such as folder structure and requirements for running python scripts
    a. Hyperlinked Trello SCRUM dashboard - Detailed breakdown of project tasks
    b. Hyperlinked GitHub repository - At least two commits and a .gitignore for dataset
2. Statement of Work - Document providing an overview of the aims and goals of this task
3. Python data preparation script - Python script containing only the data preparation steps
4. Python EDA script - Python script containing only the exploratory data analysis steps
5. Documentation - Documents procedures used in data cleaning such that it can be replicated
6. EDA report - Text report discussing insights found, conclusions, and further steps

Project Schedule

Day	Tasks
Tuesday March 10th	Choose police regions to use for analysis
Wednesday March 11th	Complete all data preparation
Thursday March 12th	Start EDA
Friday March 13th	Complete EDA and start report writeup
Monday March 16th	Complete report writeup and submit

Requirements and Acceptance Criteria

The rubric below will be used to evaluate the success of this project

Marking Scheme		
Task	**Detail**	**Marks Awarded**
Demonstrates understanding of exploratory data analysis.	Excellent analysis, including data quality assessment and descriptive and inferential statistics. The concepts covered during the module are applied correctly. Basic and additional requirements were fulfilled. Extracurricular concepts were applied.	5
	Good analysis, includes some data quality assessment, and descriptive and inferential statistics. The concepts were implemented correctly. Most of the requirements were fully covered.	3-4
	Descent analysis. Some of the concepts were implemented correctly. Majority of the activities were not fully covered or are giving incorrect answers.	1-2
Application of pandas to write concise and clear data manipulation code.	Excellent implementation and data manipulation. The code is clear and multiple elements covered during the lectures were deployed correctly. Extracurricular concepts were applied.	5
	Good implementation and data manipulation. The code is mostly clear and multiple elements covered during the lectures were deployed.	3-4
	Descent implementation. The code is sometimes difficult to follow. Some elements covered during the lectures were deployed	1-2
Appropriate visualisation methods chosen to answer questions about the data.	Excellent visualisations and graphs. Graphs are clear and supports the analysis and follow best practices. Multiple elements covered during the lectures were deployed correctly. Extracurricular concepts were applied.	5
	Good visualisations and graphs. Graphs are mostly clear and are a good tool to support the analysis following some good practices.	3-4
	Descent visualisations and graphs. Graphs are sometimes difficult to follow.	1-2
Report structure (Format, completeness, readable, coherent) and code style.	The report is complete, professional, coherent and easy to follow. Excellent implementation of coding styles and best practices in code documentation. Excellent PM documentation.	5
	In general, a good report but with some room for improvement. Good implementation of coding styles and best practices in code documentation. Good PM documentation.	3-4
	The report is incomplete. Little to none PM documentation.	1-2

Budget

There is no additional budget required for this project, all datasets and the tools used are all publicly available